

Fitting AXD

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Contents

```
dt <- as.data.table(read.csv(file.path(results_dir, "fitting_results.csv")))
# dt2 <- as.data.table(read.csv("fits_results_AXD_no_rectifier.csv"))
dt$model <- 'original'

# Number of observations per participant: n_trials (from fit_listener) equals the number of
# 'Measured' trials per participant in the behavioural data (AXD_full_dataset_20260202.csv)
dt[, N := n_trials]
dt <- dt[order(participant)]

# dt <- dt[!participant %in% c('P0204', 'P0224', 'P0228', 'P0239', 'P0244', 'P0259', 'P0330', 'P0350')]
# dt <- dt[!participant %in% c('P0204', 'P0228', 'P0239', 'P0243', 'P0244', 'P0259', 'P0330')]
# dt <- dt[!participant %in% c('P0239', 'P0243', 'P0261', 'P0020',
#                             'P0204', 'P0228', 'P0244', 'P0330')]

# dt <- dt[participant %in% n_obs_per_participant[N > 100]$participant]

# Define number of parameters per method
# You'll need to adjust these based on your actual models
# Typically: sigma_spectral, sigma_motor, sigma_prior
k_params <- 3

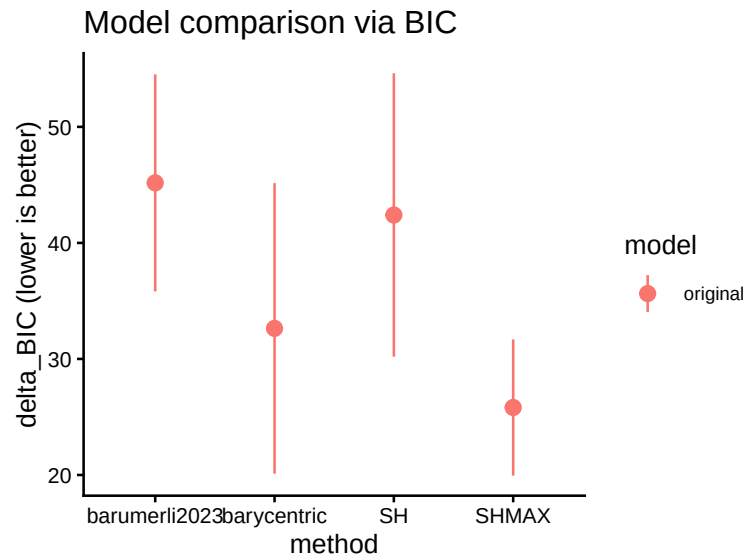
# Compute BIC
# BIC = k*log(n) + 2*NLL
dt[, BIC := k_params * log(N) + 2 * nll_500]

# For model comparison, compute delta BIC relative to baseline within each participant
dt[, delta_BIC := BIC - min(BIC), by = participant]

aa <- dcast(dt, participant ~ method, value.var = "delta_BIC")
# Count how many times each method is best
dt[, .(best_method = method[which.min(delta_BIC)]), by = participant][, .N, by = best_method]
```

```
##      best_method      N
##      <char> <int>
## 1: barycentric    13
## 2: barumerli2023   10
## 3: SHMAX          7
## 4: SH             3
```

```
# Plot delta BIC instead of llz
summarySE(dt, measurevar = "delta_BIC", groupvars = c("method", 'model')) |>
  ggplot() +
  geom_pointrange(aes(x = method, y = delta_BIC, ymin = delta_BIC-se, ymax = delta_BIC+se, color = model),
    position = position_dodge(width = .2)) +
  labs(y = "delta_BIC (lower is better)",
    title = "Model comparison via BIC")
```



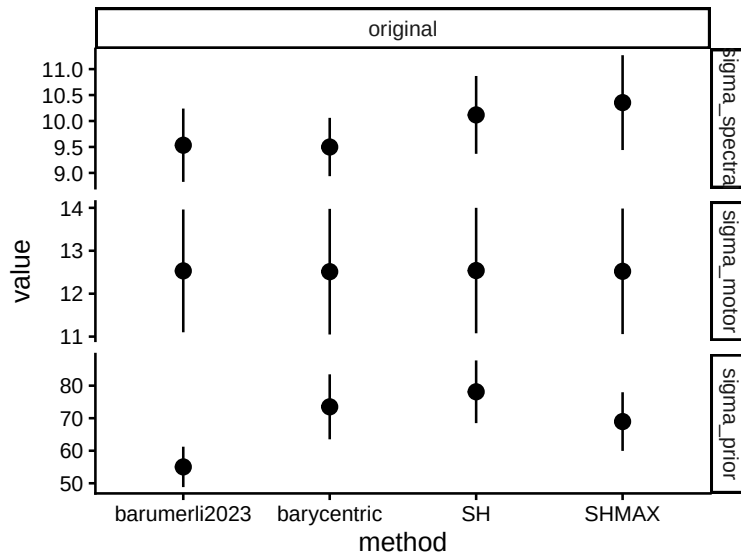
```
write.csv(dt, 'fitting_results_bic.csv')
```

```
summarySE(dt, measurevar = "delta_BIC", groupvars = c("method", 'model'))
```

```
##           method    model  N delta_BIC      sd      se      ci
## 1 barumerli2023 original  33  45.17585 53.71652  9.350845 19.04705
## 2  barycentric original  33  32.63205 71.90385 12.516854 25.49600
## 3           SH original  33  42.40493 70.15438 12.212311 24.87566
## 4          SHMAX original  33  25.81684 33.71020  5.868192 11.95312
```

```
# dt <- dt[subject != 'P0244']
```

```
dtm <- melt(dt, id.vars = c('participant', 'method', 'model'), measure.vars = c('sigma_spectral', 'sigma_spatial'))
summarySE(dtm, measurevar = "value", groupvars = c("method", 'variable', 'model')) |>
  ggplot() +
  geom_pointrange(aes(x = method, y = value, ymin = value-se, ymax = value+se))+
  facet_grid(variable ~ model, scales = "free")
```



```
dt$method <- as.factor(dt$method)
dt$participant <- as.factor(dt$participant)
am <- art(delta_BIC ~ method + Error(participant), data = dt)
anova_res <- anova(am)
anova_res
```

```
## Analysis of Variance of Aligned Rank Transformed Data
##
## Table Type: Repeated Measures Analysis of Variance Table (Type I)
## Model: Repeated Measures (aov)
## Response: art(delta_BIC)
##
##           Error Df Df.res F value  Pr(>F)
## 1 method Within  3      96  1.9054 0.13386
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# explained variance
eta_squared(anova_res)
```

```
## # Effect Size for ANOVA (Type I)
##
## Parameter | Eta2 (partial) |      95% CI
## -----
## 1          |          0.06 | [0.00, 1.00]
##
## - One-sided CIs: upper bound fixed at [1.00].
```

Posthoc

```
art.con(am, 'method')
```

```
## contrast                estimate    SE df t.ratio p.value
## barumerli2023 - barycentric    15.79  9.23  96   1.711  0.3239
## barumerli2023 - SH             -5.12  9.23  96  -0.555  0.9450
## barumerli2023 - SHMAX          6.36  9.23  96   0.689  0.9009
## barycentric - SH             -20.91  9.23  96  -2.265  0.1135
## barycentric - SHMAX          -9.42  9.23  96  -1.021  0.7375
```

```

## SH - SHMAX                11.48 9.23 96    1.244 0.6004
##
## P value adjustment: tukey method for comparing a family of 4 estimates
# =====
# 1. Friedman test
# =====
bic_wide <- dcast(dt, participant ~ method, value.var = "BIC")
friedman <- friedman.test(as.matrix(bic_wide[, .(SHMAX, SH, barycentric, barumerli2023)]))
cat("=== FRIEDMAN TEST ===\n")

## === FRIEDMAN TEST ===
print(friedman)

##
## Friedman rank sum test
##
## data: as.matrix(bic_wide[, .(SHMAX, SH, barycentric, barumerli2023)])
## Friedman chi-squared = 5.0364, df = 3, p-value = 0.1692
# =====
# 2. Pairwise Wilcoxon signed-rank tests (BIC differences)
# =====
methods <- c("SHMAX", "SH", "barycentric", "barumerli2023")
pairs <- combn(methods, 2, simplify = FALSE)

pairwise_results <- rbindlist(lapply(pairs, function(pair) {
  a <- pair[1]; b <- pair[2]
  bic_a <- dt[method == a, .(participant, BIC_a = BIC)]
  bic_b <- dt[method == b, .(participant, BIC_b = BIC)]
  merged <- merge(bic_a, bic_b, by = "participant")
  d <- merged$BIC_b - merged$BIC_a # positive = a wins

  wt <- wilcox.test(d, alternative = "two.sided", exact = FALSE)

  data.table(
    method_a = a,
    method_b = b,
    wins_a = sum(d > 0),
    wins_b = sum(d < 0),
    n = length(d),
    median_delta = median(d),
    W = as.numeric(wt$statistic),
    p = wt$p.value
  )
}))

# Holm correction
pairwise_results[, p_holm := p.adjust(p, method = "bonferroni")]

cat("\n=== PAIRWISE WILCOXON SIGNED-RANK TESTS ===\n")

##
## === PAIRWISE WILCOXON SIGNED-RANK TESTS ===

```

```
print(pairwise_results[order(p)], digits = 4)
```

```
##      method_a      method_b wins_a wins_b      n median_delta      W      p
##      <char>      <char> <int> <int> <int>      <num> <num> <num>
## 1:      SH      barycentric      11      22      33      -6.862      138 0.01117
## 2:      SHMAX barumerli2023      21      12      33      10.431      380 0.07691
## 3: barycentric barumerli2023      19      14      33      4.849      364 0.13807
## 4:      SHMAX      SH      19      14      33      2.380      349 0.22436
## 5:      SHMAX      barycentric      12      21      33      -2.732      235 0.42137
## 6:      SH barumerli2023      16      17      33      -2.434      315 0.54352
##      p_holm
##      <num>
## 1: 0.06704
## 2: 0.46145
## 3: 0.82840
## 4: 1.00000
## 5: 1.00000
## 6: 1.00000
```

```
# =====
# 3. Win counts (lowest BIC per participant)
# =====
winners <- dt[, .SD[which.min(BIC)], by = participant]

cat("\n=== WIN COUNTS (lowest BIC) ===\n")
```

```
##
## === WIN COUNTS (lowest BIC) ===
print(table(winners$method))
```

```
##
## barumerli2023      barycentric      SH      SHMAX
##              10              13              3              7
```

```
# =====
# 4. Kass & Raftery categories: SHMAX vs barumerli2023
# =====
shmax_bic <- dt[method == "SHMAX", .(participant, BIC_shmax = BIC)]
baru_bic <- dt[method == "barumerli2023", .(participant, BIC_baru = BIC)]
kr <- merge(shmax_bic, baru_bic, by = "participant")
kr[, delta := BIC_baru - BIC_shmax] # positive = SHMAX wins

cat("\n=== KASS & RAFTERY: SHMAX vs barumerli2023 ===\n")
```

```
##
## === KASS & RAFTERY: SHMAX vs barumerli2023 ===
cat(sprintf("Strong for SHMAX (dBIC > 10):      %d\n", kr[delta > 10, .N]))

## Strong for SHMAX (dBIC > 10):      17
cat(sprintf("Positive for SHMAX (2 < dBIC <= 10): %d\n", kr[delta > 2 & delta <= 10, .N]))

## Positive for SHMAX (2 < dBIC <= 10): 2
cat(sprintf("Negligible (|dBIC| <= 2):          %d\n", kr[abs(delta) <= 2, .N]))
```

```

## Negligible (|dBIC| <= 2):          4
cat(sprintf("Positive for baru (-10 <= dBIC < -2): %d\n", kr[delta < -2 & delta >= -10, .N]))

## Positive for baru (-10 <= dBIC < -2): 1
cat(sprintf("Strong for baru (dBIC < -10):      %d\n", kr[delta < -10, .N]))

## Strong for baru (dBIC < -10):      9
# =====
# 5. Kass & Raftery categories: barycentric vs barumerli2023
# =====
bary_bic <- dt[method == "barycentric", .(participant, BIC_bary = BIC)]
kr2 <- merge(bary_bic, baru_bic, by = "participant")
kr2[, delta := BIC_baru - BIC_bary]

cat("\n=== KASS & RAFTERY: barycentric vs barumerli2023 ===\n")

##
## === KASS & RAFTERY: barycentric vs barumerli2023 ===
cat(sprintf("Strong for barycentric (dBIC > 10):    %d\n", kr2[delta > 10, .N]))

## Strong for barycentric (dBIC > 10):    15
cat(sprintf("Positive for barycentric (2 < dBIC <= 10): %d\n", kr2[delta > 2 & delta <= 10, .N]))

## Positive for barycentric (2 < dBIC <= 10): 3
cat(sprintf("Negligible (|dBIC| <= 2):              %d\n", kr2[abs(delta) <= 2, .N]))

## Negligible (|dBIC| <= 2):              4
cat(sprintf("Positive for baru (-10 <= dBIC < -2):    %d\n", kr2[delta < -2 & delta >= -10, .N]))

## Positive for baru (-10 <= dBIC < -2):    0
cat(sprintf("Strong for baru (dBIC < -10):            %d\n", kr2[delta < -10, .N]))

## Strong for baru (dBIC < -10):            11
# =====
# 6. Summary table
# =====
cat("\n=== SUMMARY: dBIC from best (per participant) ===\n")

##
## === SUMMARY: dBIC from best (per participant) ===
print(dt[, .(mean = mean(delta_BIC),
  median = median(delta_BIC),
  sd = sd(delta_BIC)), by = method][order(mean)])

##          method      mean      median      sd
##          <fctr>    <num>    <num>    <num>
## 1:          SHMAX 25.81684 11.162050 33.71020
## 2:    barycentric 32.63205  3.610039 71.90385
## 3:           SH  42.40493 19.838235 70.15438
## 4: barumerli2023 45.17585 21.406172 53.71652

```

```

# =====
# 1. Motor noise consistency across methods
# =====
cat("=== MOTOR NOISE (sigma_motor) ===\n")

parameters

## === MOTOR NOISE (sigma_motor) ===
cat(sprintf("Range: %.1f-%.1f° (median = %.1f°)\n",
            dt[, min(sigma_motor)], dt[, max(sigma_motor)], dt[, median(sigma_motor)]))

## Range: 2.2-35.8° (median = 9.7°)
# Within-participant CV across methods
motor_cv <- dt[, .(sd = sd(sigma_motor), mean = mean(sigma_motor)), by = participant]
motor_cv[, cv := sd / mean]
cat(sprintf("Within-participant CV: mean = %.3f, max = %.3f\n",
            mean(motor_cv$cv), max(motor_cv$cv)))

## Within-participant CV: mean = 0.016, max = 0.115
# =====
# 2. Friedman test on sigma_spectral across methods
# =====
spec_wide <- dcast(dt, participant ~ method, value.var = "sigma_spectral")
friedman_spec <- friedman.test(as.matrix(spec_wide[, .(SHMAX, SH, barycentric, barumerli2023)]))
cat("\n=== FRIEDMAN: sigma_spectral ===\n")

##
## === FRIEDMAN: sigma_spectral ===
print(friedman_spec)

##
## Friedman rank sum test
##
## data: as.matrix(spec_wide[, .(SHMAX, SH, barycentric, barumerli2023)])
## Friedman chi-squared = 5.3323, df = 3, p-value = 0.149
# Per-method summary
cat("\n--- sigma_spectral summary by method ---\n")

##
## --- sigma_spectral summary by method ---
dt[, .(median = median(sigma_spectral),
        q25 = quantile(sigma_spectral, 0.25),
        q75 = quantile(sigma_spectral, 0.75),
        mean = mean(sigma_spectral),
        sd = sd(sigma_spectral)), by = method][order(median)] |> print(digits = 3)

##           method median  q25  q75  mean   sd
##           <fctr> <num> <num> <num> <num> <num>
## 1: barumerli2023  8.82  6.89 10.0  9.53  4.06
## 2:           SHMAX  8.96  7.64 10.6 10.35  5.25
## 3:    barycentric  9.10  7.56 10.1  9.50  3.23
## 4:           SH    9.57  7.69 11.1 10.12  4.31

```

```

# =====
# 3. Friedman test on sigma_prior across methods
# =====
prior_wide <- dcast(dt, participant ~ method, value.var = "sigma_prior")
friedman_prior <- friedman.test(as.matrix(prior_wide[, .(SHMAX, SH, barycentric, barumerli2023)]))
cat("\n=== FRIEDMAN: sigma_prior ===\n")

##
## === FRIEDMAN: sigma_prior ===

print(friedman_prior)

##
## Friedman rank sum test
##
## data: as.matrix(prior_wide[, .(SHMAX, SH, barycentric, barumerli2023)])
## Friedman chi-squared = 10.907, df = 3, p-value = 0.01224

# Per-method summary
cat("\n--- sigma_prior summary by method ---\n")

##
## --- sigma_prior summary by method ---

dt[, .(median = median(sigma_prior),
      q25 = quantile(sigma_prior, 0.25),
      q75 = quantile(sigma_prior, 0.75),
      mean = mean(sigma_prior),
      sd = sd(sigma_prior)), by = method][order(median)] |> print(digits = 3)

##
##      method median  q25  q75  mean   sd
##      <fctr>  <num> <num> <num> <num> <num>
## 1:      SHMAX   50.0  32.8  88.8  69.0  51.8
## 2:  barycentric   50.0  35.6 121.3  73.5  57.4
## 3: barumerli2023   50.0  29.9  73.5  55.0  35.6
## 4:           SH   50.5  40.3 129.9  78.1  55.4

# Pairwise Wilcoxon on sigma_prior (follow-up to significant Friedman)
cat("\n--- sigma_prior pairwise Wilcoxon ---\n")

##
## --- sigma_prior pairwise Wilcoxon ---

methods <- c("SHMAX", "SH", "barycentric", "barumerli2023")
pairs <- combn(methods, 2, simplify = FALSE)
prior_pairwise <- rbindlist(lapply(pairs, function(pair) {
  a <- prior_wide[[pair[1]]]
  b <- prior_wide[[pair[2]]]
  wt <- wilcox.test(a, b, paired = TRUE, exact = FALSE)
  data.table(method_a = pair[1], method_b = pair[2],
             median_a = median(a), median_b = median(b),
             V = as.numeric(wt$statistic), p = wt$p.value)
}))
prior_pairwise[, p_holm := p.adjust(p, method = "holm")]
print(prior_pairwise[order(p)], digits = 4)

##      method_a      method_b median_a median_b      V      p      p_holm

```

```
##          <char>          <char>      <num>      <num> <num>      <num>      <num>
## 1: barycentric barumerli2023    50.00    50.00   414 0.005182 0.03109
## 2:          SH barumerli2023    50.53    50.00   413 0.005490 0.03109
## 3:          SHMAX barumerli2023  50.00    50.00   417 0.015096 0.06038
## 4:          SHMAX barycentric   50.00    50.00   196 0.649766 1.00000
## 5:          SHMAX          SH   50.00    50.53   240 0.660354 1.00000
## 6:          SH barycentric     50.53    50.00   272 0.888469 1.00000
```

```
# =====
# 4. Cross-method Spearman correlations
# =====
cat("\n=== CROSS-METHOD CORRELATIONS: sigma_spectral ===\n")
```

```
##
## === CROSS-METHOD CORRELATIONS: sigma_spectral ===
for (i in 1:(length(methods) - 1)) {
  for (j in (i + 1):length(methods)) {
    ct <- cor.test(spec_wide[[methods[i]]], spec_wide[[methods[j]]], method = "spearman")
    cat(sprintf(" %s vs %s: rho = %.3f, p = %.4f\n",
                methods[i], methods[j], ct$estimate, ct$p.value))
  }
}
```

```
## SHMAX vs SH: rho = 0.711, p = 0.0000
## SHMAX vs barycentric: rho = 0.809, p = 0.0000
## SHMAX vs barumerli2023: rho = 0.453, p = 0.0081
## SH vs barycentric: rho = 0.728, p = 0.0000
## SH vs barumerli2023: rho = 0.569, p = 0.0005
## barycentric vs barumerli2023: rho = 0.484, p = 0.0043
cat("\n=== CROSS-METHOD CORRELATIONS: sigma_prior ===\n")
```

```
##
## === CROSS-METHOD CORRELATIONS: sigma_prior ===
for (i in 1:(length(methods) - 1)) {
  for (j in (i + 1):length(methods)) {
    ct <- cor.test(prior_wide[[methods[i]]], prior_wide[[methods[j]]], method = "spearman")
    cat(sprintf(" %s vs %s: rho = %.3f, p = %.4f\n",
                methods[i], methods[j], ct$estimate, ct$p.value))
  }
}
```

```
## SHMAX vs SH: rho = 0.790, p = 0.0000
## SHMAX vs barycentric: rho = 0.897, p = 0.0000
## SHMAX vs barumerli2023: rho = 0.821, p = 0.0000
## SH vs barycentric: rho = 0.726, p = 0.0000
## SH vs barumerli2023: rho = 0.733, p = 0.0000
## barycentric vs barumerli2023: rho = 0.828, p = 0.0000
```

```

# =====
# 5. sigma_spectral vs sigma_prior correlation (per method)
# =====
cat("\n=== sigma_spectral vs sigma_prior WITHIN METHOD ===\n")

##
## === sigma_spectral vs sigma_prior WITHIN METHOD ===
for (m in methods) {
  sub <- dt[method == m]
  ct <- cor.test(sub$sigma_spectral, sub$sigma_prior, method = "spearman")
  cat(sprintf(" %s: rho = %.3f, p = %.4f\n", m, ct$estimate, ct$p.value))
}

## SHMAX: rho = -0.334, p = 0.0573
## SH: rho = -0.168, p = 0.3493
## barycentric: rho = -0.251, p = 0.1580
## barumerli2023: rho = -0.242, p = 0.1753

# =====
# 6. Within-participant variability of sigma_spectral and sigma_prior
# =====
cat("\n=== WITHIN-PARTICIPANT RANGE ACROSS METHODS ===\n")

##
## === WITHIN-PARTICIPANT RANGE ACROSS METHODS ===
spec_range <- dt[, .(range = max(sigma_spectral) - min(sigma_spectral)), by = participant]
cat(sprintf("sigma_spectral: median range = %.1f, mean range = %.1f\n",
  median(spec_range$range), mean(spec_range$range)))

## sigma_spectral: median range = 3.1, mean range = 4.4
prior_range <- dt[, .(range = max(sigma_prior) - min(sigma_prior)), by = participant]
cat(sprintf("sigma_prior: median range = %.1f°, mean range = %.1f°\n",
  median(prior_range$range), mean(prior_range$range)))

## sigma_prior: median range = 22.9°, mean range = 44.0°

# =====
# 7. Overall summary for reporting
# =====
cat("\n=== OVERALL SUMMARY (all methods pooled) ===\n")

##
## === OVERALL SUMMARY (all methods pooled) ===
cat(sprintf("sigma_motor: %.1f-%.1f° (median %.1f°)\n",
  dt[, min(sigma_motor)], dt[, max(sigma_motor)], dt[, median(sigma_motor)]))

## sigma_motor: 2.2-35.8° (median 9.7°)
cat(sprintf("sigma_spectral: %.1f-%.1f (median %.1f)\n",
  dt[, min(sigma_spectral)], dt[, max(sigma_spectral)], dt[, median(sigma_spectral)]))

## sigma_spectral: 3.2-29.6 (median 9.0)

```

```
cat(sprintf("sigma_prior:    %.1f-%.1f° (median %.1f°)\n",
           dt[, min(sigma_prior)], dt[, max(sigma_prior)], dt[, median(sigma_prior)]))
```

```
## sigma_prior:    4.4-179.9° (median 50.0°)
```

posterior

```
df <- as.data.table(read.csv(file.path(results_dir, "synthetic_vs_actual_comparison.csv")))
```

```
actual <- df[data_type == "actual"]
setkey(actual, participant)
```

```
methods <- c("SHMAX", "barycentric", "SH", "barumerli2023")
metrics <- c("rmsL_deg", "rmsPmedianlocal_deg", "querrMiddlebrooks")
metric_labels <- c("rmsL", "PE (local)", "QE (%)")
```

```
# =====
# 1. Correlations and bias for main metrics
# =====
cat("=== POSTERIOR PREDICTIVE CHECKS ===\n\n")
```

```
## === POSTERIOR PREDICTIVE CHECKS ===
```

```
results <- rbindlist(lapply(methods, function(m) {
  model <- df[method == m]
  setkey(model, participant)
  common <- intersect(actual$participant, model$participant)

  rbindlist(lapply(seq_along(metrics), function(i) {
    a <- actual[common, get(metrics[i])]
    mod <- model[common, get(metrics[i])]

    ct <- cor.test(a, mod, method = "pearson")
    wt <- wilcox.test(mod - a, exact = FALSE)

    data.table(
      method = m,
      metric = metric_labels[i],
      mean_actual = mean(a),
      mean_model = mean(mod),
      bias = mean(mod) - mean(a),
      r = ct$estimate,
      r_p = ct$p.value,
      W = as.numeric(wt$statistic),
      wilcox_p = wt$p.value
    )
  })))
}))

cat("--- Pearson r ---\n")
```

```
## --- Pearson r ---
```

```
r_wide <- dcast(results, metric ~ method, value.var = "r")
print(r_wide, digits = 3)
```

```
## Key: <metric>
##      metric      SH SHMAX barumerli2023 barycentric
##      <char> <num> <num>      <num>      <num>
## 1: PE (local) 0.569 0.634      0.651      0.648
## 2:   QE (%) 0.515 0.524      0.186      0.601
## 3:   rmsL 0.892 0.893      0.891      0.895

cat("\n--- Pearson r p-values ---\n")

##
## --- Pearson r p-values ---
rp_wide <- dcast(results, metric ~ method, value.var = "r_p")
print(rp_wide, digits = 4)

## Key: <metric>
##      metric      SH      SHMAX barumerli2023 barycentric
##      <char>      <num>      <num>      <num>      <num>
## 1: PE (local) 5.565e-04 7.476e-05      4.156e-05 4.553e-05
## 2:   QE (%) 2.179e-03 1.733e-03      2.988e-01 2.171e-04
## 3:   rmsL 3.179e-12 2.883e-12      3.846e-12 2.267e-12

cat("\n--- Mean bias (model - actual) ---\n")

##
## --- Mean bias (model - actual) ---
bias_wide <- dcast(results, metric ~ method, value.var = "bias")
print(bias_wide, digits = 2)

## Key: <metric>
##      metric      SH  SHMAX barumerli2023 barycentric
##      <char> <num>  <num>      <num>      <num>
## 1: PE (local) 0.14 -0.054      0.51      -0.36
## 2:   QE (%) 1.27 1.318      2.30      0.46
## 3:   rmsL 0.74 0.590      0.90      0.61

cat("\n--- Wilcoxon signed-rank p (model vs actual) ---\n")

##
## --- Wilcoxon signed-rank p (model vs actual) ---
wp_wide <- dcast(results, metric ~ method, value.var = "wilcox_p")
print(wp_wide, digits = 4)

## Key: <metric>
##      metric      SH  SHMAX barumerli2023 barycentric
##      <char> <num>  <num>      <num>      <num>
## 1: PE (local) 0.9430 0.6422      0.7342      0.4423
## 2:   QE (%) 0.3528 0.2679      0.1802      0.5085
## 3:   rmsL 0.2603 0.4111      0.1288      0.4111

# =====
# 2. Actual data summary
# =====
cat("\n=== ACTUAL DATA SUMMARY ===\n")

##
## === ACTUAL DATA SUMMARY ===
```

```

n <- actual[, .N]
for (i in seq_along(metrics)) {
  vals <- actual[, get(metrics[i])]
  cat(sprintf("%s: %.1f ± %.1f (SE), SD = %.1f, N = %d\n",
              metric_labels[i], mean(vals), sd(vals) / sqrt(n), sd(vals), n))
}

```

```

## rmsL: 12.7 ± 0.9 (SE), SD = 5.2, N = 33
## PE (local): 33.8 ± 1.1 (SE), SD = 6.2, N = 33
## QE (%): 15.1 ± 1.8 (SE), SD = 10.6, N = 33

```

```

# =====
# 3. Accuracy metrics (accL, accP) - light treatment
# =====
cat("\n=== ACCURACY METRICS (accL, accP at ±30°) ===\n\n")

```

```

##
## === ACCURACY METRICS (accL, accP at ±30°) ===

acc_metrics <- c("accL_cutoff_30_deg", "accP_cutoff_30_deg")
acc_labels <- c("accL (±30°)", "accP (±30°)")

```

```

acc_results <- rbindlist(lapply(methods, function(m) {
  model <- df[method == m]
  setkey(model, participant)
  common <- intersect(actual$participant, model$participant)

  rbindlist(lapply(seq_along(acc_metrics), function(i) {
    a <- actual[common, get(acc_metrics[i])]
    mod <- model[common, get(acc_metrics[i])]

    ct <- cor.test(a, mod, method = "pearson")
    wt <- wilcox.test(mod - a, exact = FALSE)

    data.table(
      method = m,
      metric = acc_labels[i],
      mean_actual = mean(a),
      mean_model = mean(mod),
      bias = mean(mod) - mean(a),
      r = ct$estimate,
      r_p = ct$p.value,
      W = as.numeric(wt$statistic),
      wilcox_p = wt$p.value
    )
  })))

```

```

print(acc_results[, .(method, metric, mean_actual = round(mean_actual, 2),
                    mean_model = round(mean_model, 2),
                    bias = round(bias, 2),
                    r = round(r, 3), r_p = round(r_p, 4),
                    wilcox_p = round(wilcox_p, 4))])

```

```

##           method      metric mean_actual mean_model  bias      r      r_p wilcox_p

```

##	<char>	<char>	<num>	<num>	<num>	<num>	<num>	<num>
## 1:	SHMAX	accL ($\pm 30^\circ$)	3.17	0.00	-3.17	0.219	0.2211	0.0032
## 2:	SHMAX	accP ($\pm 30^\circ$)	0.60	1.63	1.03	0.222	0.2138	0.5435
## 3:	barycentric	accL ($\pm 30^\circ$)	3.17	-0.02	-3.20	0.190	0.2895	0.0034
## 4:	barycentric	accP ($\pm 30^\circ$)	0.60	1.75	1.15	0.110	0.5411	0.5317
## 5:	SH	accL ($\pm 30^\circ$)	3.17	0.05	-3.13	0.186	0.3004	0.0036
## 6:	SH	accP ($\pm 30^\circ$)	0.60	1.39	0.79	0.040	0.8263	0.6811
## 7:	barumerli2023	accL ($\pm 30^\circ$)	3.17	-0.03	-3.21	0.195	0.2761	0.0034
## 8:	barumerli2023	accP ($\pm 30^\circ$)	0.60	2.60	2.00	0.057	0.7538	0.2603